O-A-429D
July 3, 1984
SUPERSEDING
O-A-429C
April 27, 1973

FEDERAL SPECIFICATION

ALUMINUM SULFATE, TECHNICAL

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types and three classes of technical grade hydrated aluminum sulfate (sulfate of alumina).

1.2 Classification. Aluminum sulfate shall be of the following types and classes as specified (see 6.2):

Type I - Purified
Type II - Unpurified

Class 1 - Lump
Class 2 - Ground
Class 3 - Rice ground

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

NN-P-71 - Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
UU-S-48 - Sacks, Shipping, Paper

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies)

FSC 6810

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(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM Standards:

D 1193 - Reagent Water
D 3951 - Commercial Packaging
E 11 - Wire-Cloth Sieves for Testing Purposes

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Chemical characteristics. Aluminum sulfate shall conform to the chemical characteristics of table I when tested as specified therein.

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TABLE I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Type I	Type II	
Iron, as Fe_2O_3, maximum	0.75	0.75	4.2.4.1
Water-soluble alumina, as Al_2O_3, minimum	17.0	17.0	4.2.4.2
Water insoluble matter, maximum	0.5	10.0	4.2.4.3
Basicity, minimum	0.05	0.05	4.2.4.4

3.2 Particle size characteristics. Aluminum sulfate shall conform to the particle size characteristics of table II when tested as specified in 4.2.4.5.

TABLE II. Particle size characteristics

Characteristic	Percent by weight		
	Class 1	Class 2	Class 3
Retained on a 75-mm sieve, maximum	0	---	---
Passing a 19-mm sieve, maximum	0	---	---
Retained on a 4.75-mm sieve, maximum	---	0.0	0.0
Retained on a 2.00-mm sieve, maximum	---	10.0	---
Retained on a 850-micrometer sieve, minimum	---	---	80.0
Passing a 150-micrometer sieve, maximum	---	---	2.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the aluminum sulfate of one type and class, produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table III. A representative specimen of approximately 2.8 kilograms (kg) for class 1 aluminum sulfate or 0.5 kg for class 2 or 3 aluminum sulfate shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE III. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container, ready for shipment. Sample shipping containers shall be examined for the following defects using an AQL of 1.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing or illegible
(f) Unitization not as specified

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Unless otherwise specified, all weighings shall be accurate to the nearest 0.1 milligram (mg). Tests shall be conducted as follows:

4.2.4.1 Iron. Dissolve approximately 10 grams (g) of specimen in 100 milliliters (mL) of hot water. After stirring for a few minutes, digest on a hotplate for 30 minutes just below the boiling point. Filter through Whatman

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No. 42 filter paper or equal. Wash with hot water until a washing is free of sulfate (approximately 10 separate washings). Collect the filtrate and washings in a 250-mL Erlenmeyer flask and close with a stopper fitted with a Bunsen valve. Add approximately 3 g of granulated zinc and 8 to 10 mL of concentrated sulfuric acid. After the zinc has dissolved, boil the solution for 2 to 3 minutes. Cool to 20° to 30°C under running tap water. Add 40 mL of 1 to 3 phosphoric acid and 4 drops of sodium diphenylamine sulfonate indicator having a concentration of 2 g per liter. Titrate slowly with 0.1N potassium dichromate solution, while stirring constantly, until the color of the solution changes from green to gray-green. Continue the titration by drops until the first tinge of purple or violet-blue appears. Run a blank determination omitting the specimen. Calculate the percent by weight iron, as Fe_2O_3, as follows:

$$\text{Percent iron} = \frac{7.985A\ (B - C + 0.05)}{W}$$

where: A = Normality of the potassium dichromate solution used,
B = Milliliters of potassium dichromate solution used to titrate the specimen,
C = Milliliters of potassium dichromate solution used to titrate the blank, and
W = Weight of specimen in grams.

4.2.4.2 Water-soluble alumina. Dissolve approximately 5 g of specimen in 100 mL of water and digest on a steam bath for 5 minutes. Stir and filter hot through a tared Gooch crucible. Wash with hot water until a washing is free of sulfates (approximately 10 separate washings). Retain the residue for the water insoluble matter determination in 4.2.4.3. Transfer the filtrate and washings to a 500-mL volumetric flask, cool to room temperature, and make up to volume with water. Transfer a 50-mL aliquot to a 400-mL beaker and dilute with 150 mL of water. Add 10 mL of concentrated hydrochloric acid and a few drops of concentrated nitric acid and heat to boiling. Add 3 drops of methyl red indicator (1 g in 500 mL of 95-percent ethanol) and a small amount of ashless filter paper pulp. Neutralize with concentrated ammonium hydroxide solution and add 3 drops in excess. Boil for 2 minutes. Allow the precipitate to settle and filter quantitatively through Whatman No. 41 filter paper or equal. Return the filter paper and contents to the 400-mL beaker, add 10 mL of concentrated hydrochloric acid, and macerate the paper to a pulp with the aid of a stirring rod. Add 150 mL of hot water and 3 drops of the methyl red indicator. Heat to boiling. Neutralize with concentrated ammonium hydroxide solution and add 3 drops in excess. Boil for 2 minutes. Allow the precipitate to settle and quantitatively filter through Whatman No. 41 filter paper or equal. Wash the filter six times with hot 2-percent ammonium nitrate solution. Ignite the filter paper and precipitate in a tared crucible at 1200°C to a constant weight. Calculate the percent by weight water-soluble alumina as follows:

$$\text{Percent water-soluble alumina} = \frac{100A}{B} - C$$

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where: A = Weight of residue in grams,
B = Weight of specimen in aliquot in grams, and
C = Percent iron determined in 4.2.4.1.

4.2.4.3 Water insoluble matter. Dry the Gooch crucible and residue retained from 4.2.4.2 at 100°C to a constant weight. Cool and weigh. Calculate the percent by weight water insoluble matter as follows:

$$\text{Percent water insoluble matter} = \frac{100A}{B}$$

where: A = Weight of insoluble matter in grams, and
B = Weight of specimen in 4.2.4.2 in grams.

4.2.4.4 Basicity.

(a) Potassium fluoride solution. Dissolve 1,000 g of potassium fluoride in 1,200 mL of hot water (carbon dioxide-free). Add 0.5 mL of phenolphthalein solution (0.1 percent in ethanol) and neutralize with 0.5N potassium hydroxide solution or 0.5N sulfuric acid until 1 mL of solution in 10 mL of carbon dioxide-free water shows a faint pink color. Filter out any insoluble matter without washing. Transfer the filtrate quantitatively to a 2,000-mL volumetric flask and make up to a volume with carbon dioxide-free water. (Note. If stored, keep in a plastic bottle or in a bottle coated inside with wax.)

(b) 0.5N potassium hydroxide solution. Standardize the potassium hydroxide solution against 0.5N sulfuric acid in approximately 40 mL of water to which 10 mL of potassium fluoride solution prepared as specified in (a) has been added. Use 0.5 mL of 1-percent phenolphthalein in ethanol as the indicator.

(c) Procedure. Dissolve 3.5 g of specimen in water and heat to boiling. Pipet 10 mL of 0.5N sulfuric acid into the hot solution and cool to room temperature. Add 18 to 20 mL of potassium fluoride solution prepared as specified in (a) and 0.5 mL of 0.1 percent phenolphthalein in ethanol to the cooled solution. Titrate with 0.5N potassium hydroxide solution prepared as specified in (b) dropwise until a slight pink color persists for 1 minute. Calculate the percent by weight basicity, as Al_2O_3, as follows:

$$\text{Percent basicity} = \frac{1.7(AB - CD)}{W}$$

where: A = Milliliters of sulfuric acid added,
B = Normality of the sulfuric acid,
C = Milliliters of potassium hydroxide solution used in the titration,
D = Normality of the potassium hydroxide solution, and
W = Weight of specimen in grams.

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4.2.4.5 Particle size. Tare and nest appropriate size sieves conforming to ASTM E 11 in descending order (coarsest sieve on top) on a receiving pan. Weigh to the nearest 0.1 g 1,000 ± 50 g of class 1 specimen or 100.0 ± 0.1 g of class 2 or 3 specimen onto the top sieve, cover, and shake for 5 minutes on a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of striker per minute. Weigh the amount of material retained on each sieve and the amount in the receiving pan to the nearest 0.1 g and calculate the percent by weight retained on or passing through each sieve, as applicable.

5. PACKAGING

5.1 Unit packing. Aluminum sulfate shall be unit packed level A, B or commercial as specified (see 6.2).

5.1.1 Level A. A quantity of 100 (+1 or -0) pounds (lb) of aluminum sulfate shall be unit packed in a multi-wall paper shipping sack conforming to type optional, style B, construction number 15 or 15X, moisture barrier grade MB2 of UU-S-48. The sack shall be closed as specified in the appendix to UU-S-48.

5.1.2 Level B. A quantity of 100 (+1 or -0) lb of aluminum sulfate shall be unit packed level B in the same manner as specified for level A in 5.1.1, except that the sack shall conform to type optional, style B, construction number 14 or 14X, moisture barrier grade MB2 of UU-S-48.

5.1.3 Commercial. A quantity of 100 (+1 or -0) lb of aluminum sulfate shall be unit packed commercially in accordance with ASTM D 3951.

5.2 Packing. Aluminum sulfate, unit packed as specified in 5.1 above, shall require no further protection for shipment other than unitization.

5.2.1 Unitization. Level A and B packs of aluminum sulfate shall be palletized in accordance with load type XV, using fiberboard cap storage aid 4; bonding means K plus 2, O, or P of MIL-STD-147. The pallet shall conform to type IV softwood of NN-P-71. Commercial packs shall be unitized in a manner to assure carrier acceptance and safe delivery from supply source to initial destination and stability in storage for a minimum duration of six months.

5.4 Marking. For civil agencies, level A and B packs and pallet loads shall be marked in accordance with Fed. Std. No. 123. For military activities, level A and B packs and pallet loads shall be marked in accordance with MIL-STD-129. Commercial packs shall be marked in accordance with ASTM D 3951. Pallet loads of commercial packs shall be marked to assure identification of their load and prompt delivery to destination. Packs shall be marked to show lot or batch number and date of manufacture of the aluminum sulfate.

6. NOTES

6.1 Intended use. Aluminum sulfate is intended for use in the treatment of water and sewage.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Type and class of aluminum sulfate required (see 1.2), and
(c) Level of unit packing required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

MILITARY INTERESTS:

Custodians:

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Air Force - 68

Review activities:

Army - MD, ME
DLA - GS

User activity:

Navy - YD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
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Preparing Activity:

Army - EA

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